Aman Prajapati

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SUMMARY

Final-year B.E. ECE student with hands-on multi-domain robotics and embedded internship experience across four organizations. Currently an Electronics & Programming Intern at Technobotix Pvt Ltd, working on Raspberry Pi–based systems, ESC configuration, RC transmitter-receiver integration, and PCB panelization. Previously built IoT sensor pipelines, OpenCV computer vision workflows, and n8n automation as a Robotics Intern at Botmakers. Skilled in ESP32/Arduino embedded systems, KiCad PCB design, GPS & ultrasonic sensor bring-up, and Python/C++ programming. National finalist, Smart India Hackathon 2025. Based in Pune — seeking a 12-month on-site Robotics Internship.

SKILLS

Robotics & ROS: ROS/ROS2 basics; sensor integration (GPS/NEO-M8N, simulation-to-reality mindset); RC systems — FlySky transmitter/receiver configuration; ESC (Electronic Speed Controller) tuning via ESC configurator

Computer Vision: OpenCV — real-time object detection, image processing, vision-to-control integration

Programming: Python, C, C++ (firmware & robotics); Git/GitHub; Linux (Ubuntu) basics

Embedded & MCU: ESP32 (PICO-D4), Arduino Uno, Raspberry Pi; UART, SPI, I2C; IoT sensor pipelines

PCB & Hardware: KiCad (multi-layer layout, DRC, Gerbers); PCB panelizing; soldering; multimeter, oscilloscope

Automation: n8n workflow automation; VS Code

INTERNSHIP EXPERIENCE

Electronics & Programming Intern | *Technobotix Pvt Ltd* **Jul 2026 – Present**

- Onboarded onto Raspberry Pi–based embedded platforms for robotics applications, building foundational skills in single-board computer setup and interfacing.
- Configured and tuned Electronic Speed Controllers (ESCs) using ESC configurator software for motor control.
- Worked with FlySky transmitter–receiver systems for RC/robot control interfacing and signal calibration.
- Executed PCB panelizing for multi-board layouts to streamline manufacturing and assembly workflows.

Robotics Intern | *Botmakers Pvt Ltd* **Mar 2026 – Jun 2026**

- Integrated IoT sensors into robotic platforms — configuring hardware interfaces, validating sensor data, and debugging cross-board signal faults.
- Built OpenCV computer vision pipelines for real-time object detection; integrated perception outputs into robot control workflows.
- Automated multi-system data flows between sensors, APIs, and robotic subsystems using n8n; documented configurations and test results systematically.

Python Programming Intern | *Oasis Infobyte* **1 Jan 2026 – 5 Feb 2026**

- Completed a one-month internship focused on hands-on Python programming, strengthening core concepts through practical task-based learning.

Intern – Electronics / PCB Design | *Javak Microcircuits Pvt Ltd* **1 Jan 2026 – 31 Jan 2026**

- Worked on PCB design and manufacturing processes during the internship period.
- Gained practical exposure to real-world electronics and hardware development workflows.
- Strengthened understanding of PCB design practices and fabrication considerations.

PROJECTS

Smart AirTag (Custom PCB Design & Prototyping) https://github.com/AmanGit08/Tagonce.git

- Designed and prototyped a compact, real-time tracking device by executing the multi-layer PCB layout in KiCad, integrating the ESP32-PICO-D4 and NEO-M8N GPS. Implemented an IoT framework for continuous, remote location tracking (SIH Project).

AI Voice Assistant (Jarvis) https://github.com/AmanGit08/Jarvis

- Built a Python-based AI voice assistant using speech recognition and the OpenAI API to process natural language commands; can open web platforms such as Google, LinkedIn, and YouTube.

Parking Management System https://github.com/AmanGit08/parkingmange.git

- Developed a parking management system that determines available parking slots by measuring distance using ultrasonic sensors and processing the data with an Arduino Uno microcontroller.

SPECIAL ACHIEVEMENT

Finalist – Smart India Hackathon (SIH) 2025 | Team Ataltra

Selected nationally for SenseStamp — a blockchain + IoT food traceability solution; recognised for hardware-software integration and technical problem solving.

Avishkar State-Level Competition | Zonal Qualifier (×2) | Technical Head, Nakshatra 4.0 Fest

EDUCATION

B.E. – Electronics & Telecommunication Engineering | *Dr. D Y Patil Inst. of Engg., Mgt. & Research, Akurdi* **Nov 2023 – Jun 2027**

CGPA: 7.78/10 (up to 6th semester)